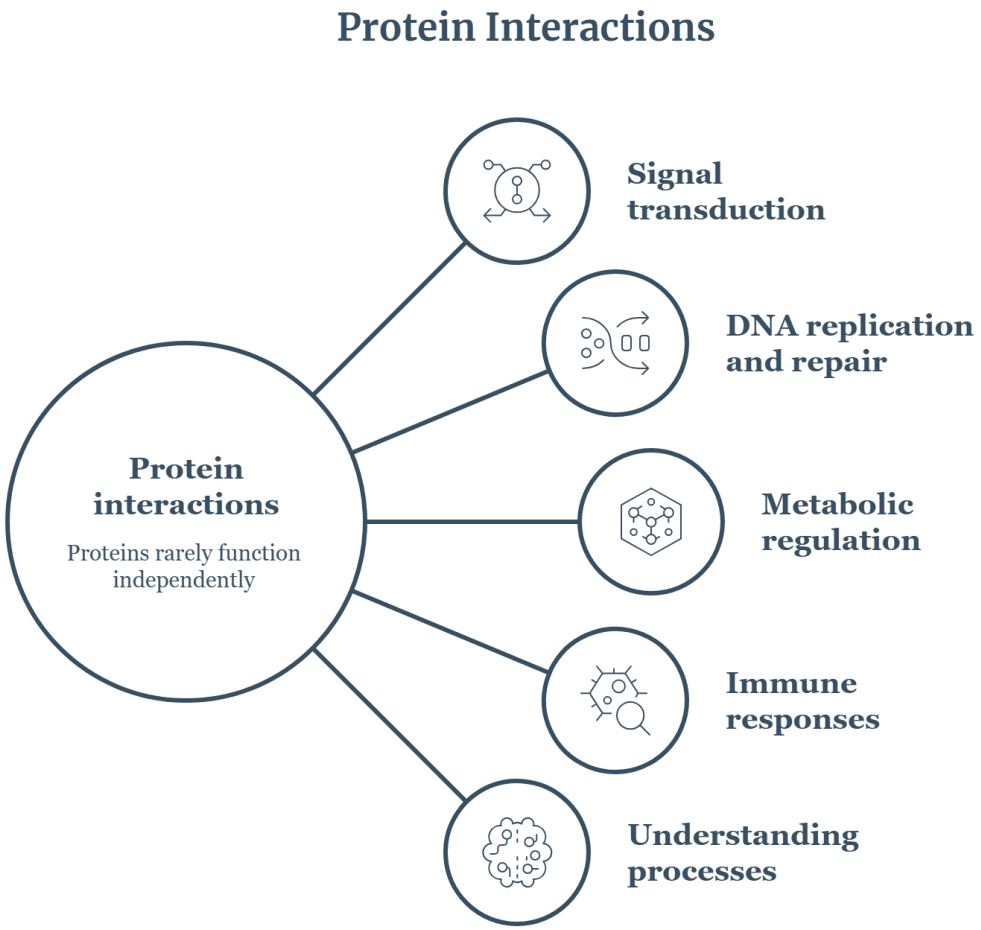
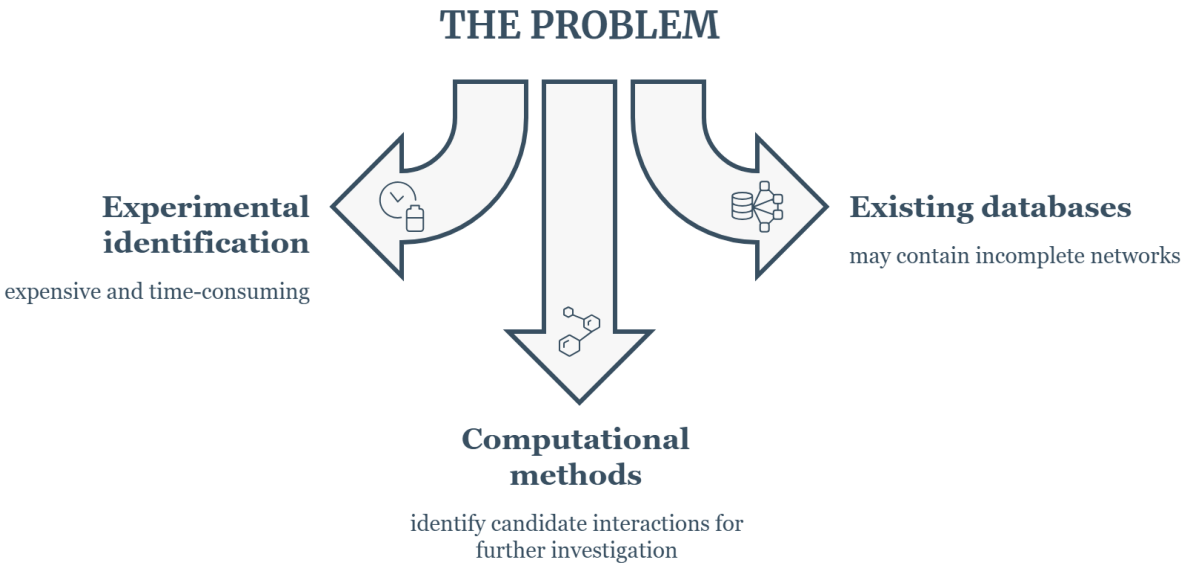
Predicting Protein–Protein Interactions Using node2vec

Evaluating Network Embeddings Under Different Negative-Sampling Strategies

Biological Background and Motivation



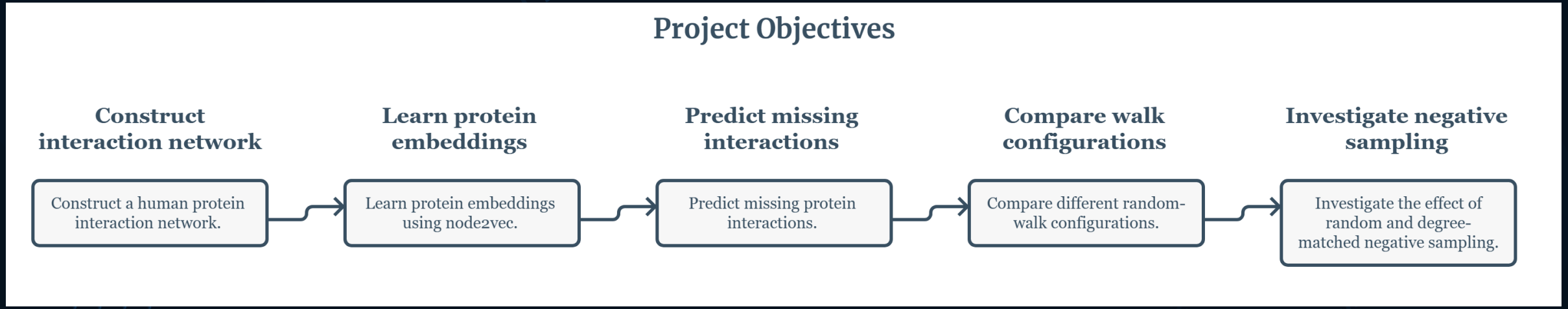


Research Question

Can node2vec learn useful representations of proteins from a protein–protein interaction network and use these representations to predict missing interactions?

Additional question introduced in my project:

- How does negative-sampling strategy affect the performance of node2vec compared with a Common Neighbors baseline?



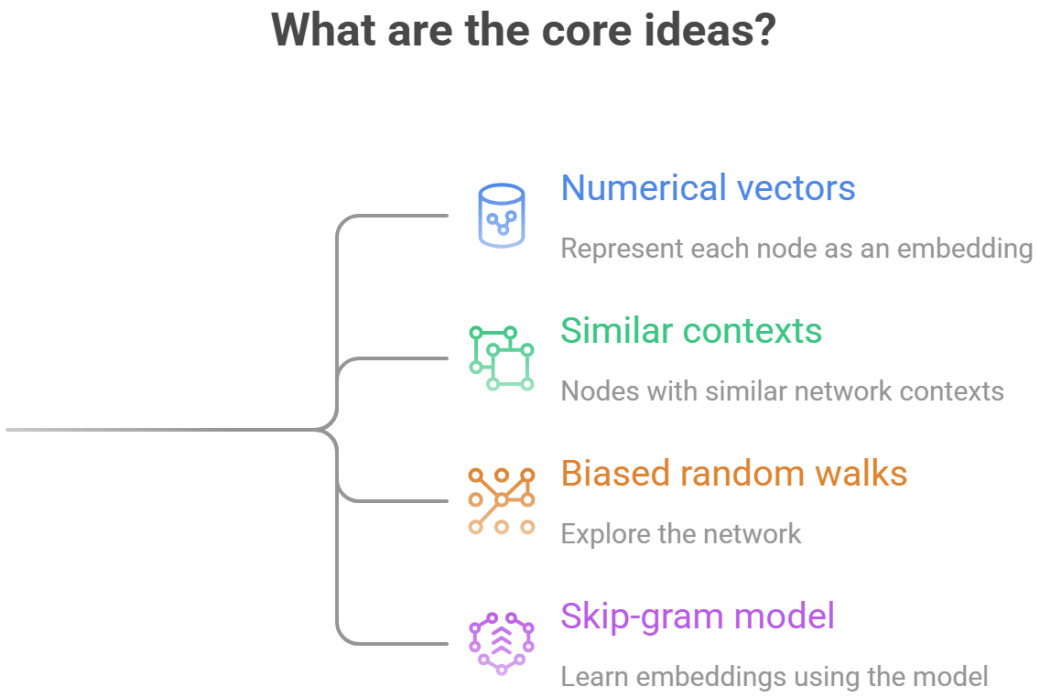
Primary Research Paper: node2vec

Reference:

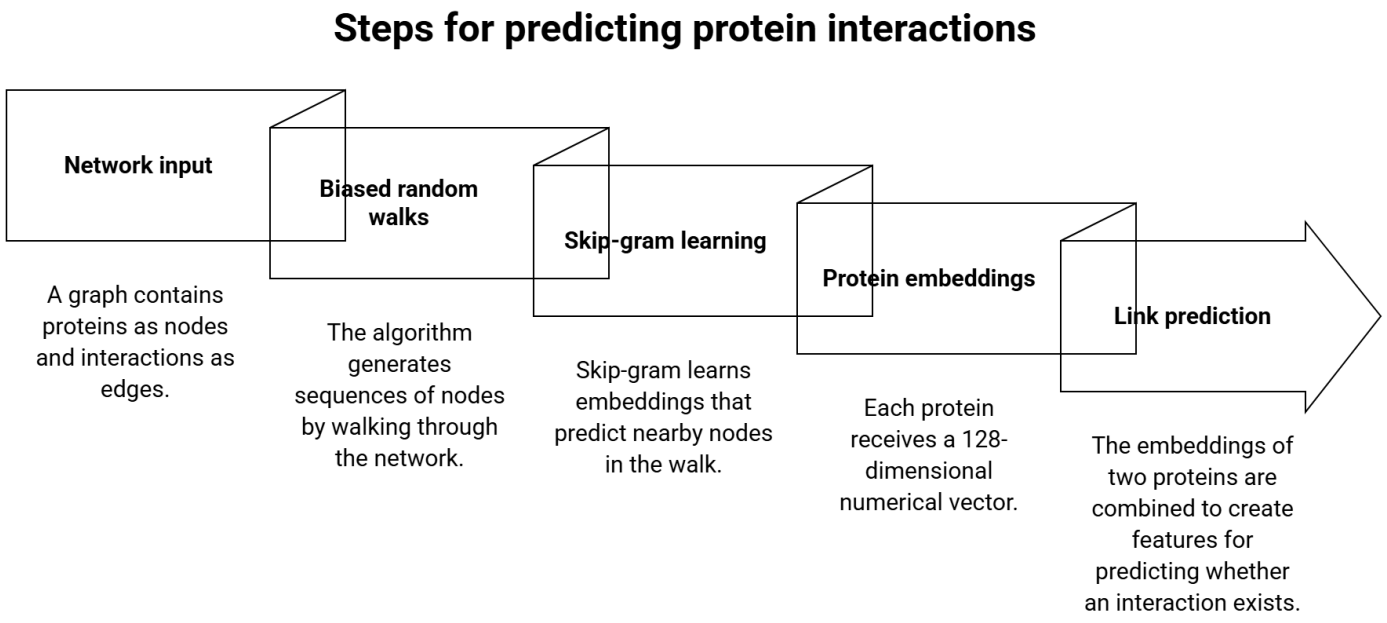
Grover and Leskovec (2016), node2vec: Scalable Feature Learning for Networks, KDD 2016.

Question addressed by the original paper:

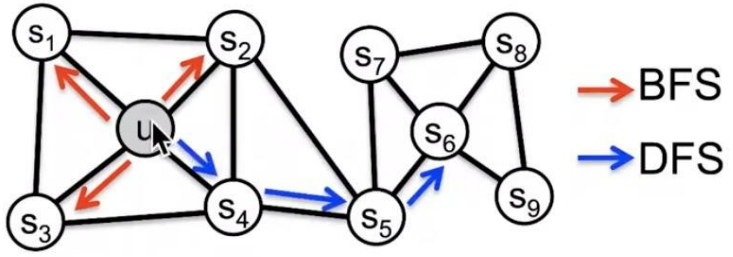
How can we learn low-dimensional representations of network nodes that preserve important network relationships?



How node2vec Works



Idea: use flexible, biased random walks that can trade off between local and global views of the network (Grover and Leskovec, 2016).



Dataset and Network Construction

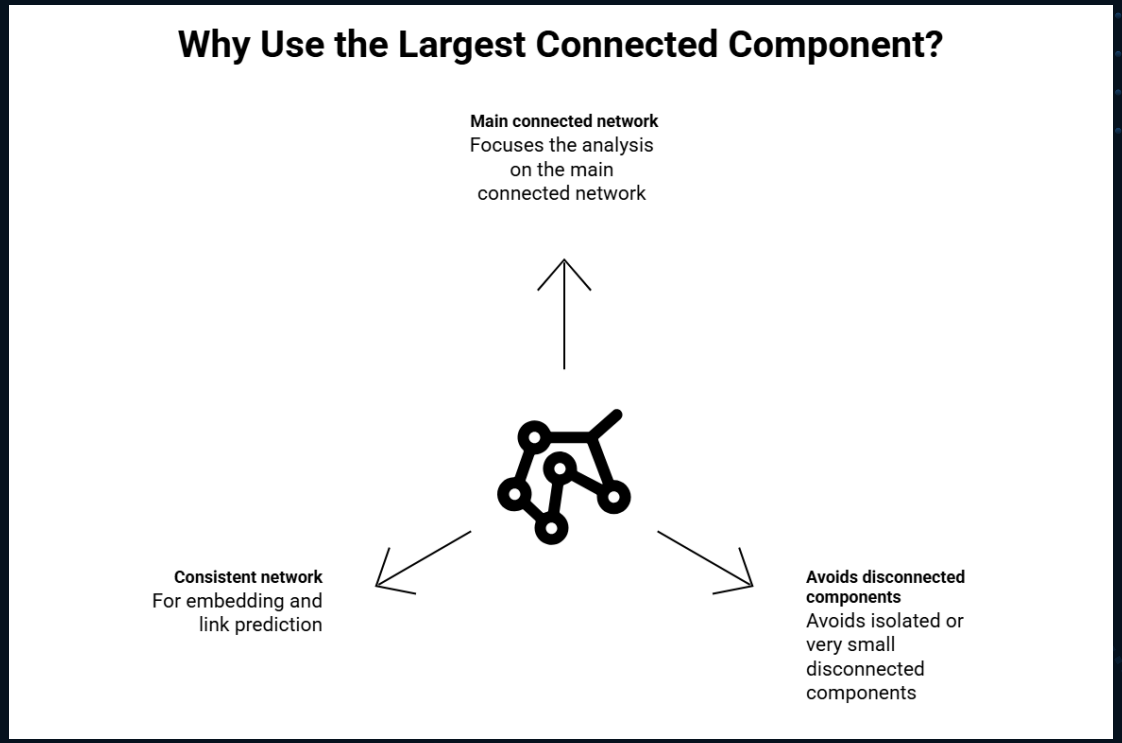
Data source: STRING v12.5, Organism: Homo sapiens (9606).

Interaction type: Physical protein interactions.

Confidence threshold: 700.

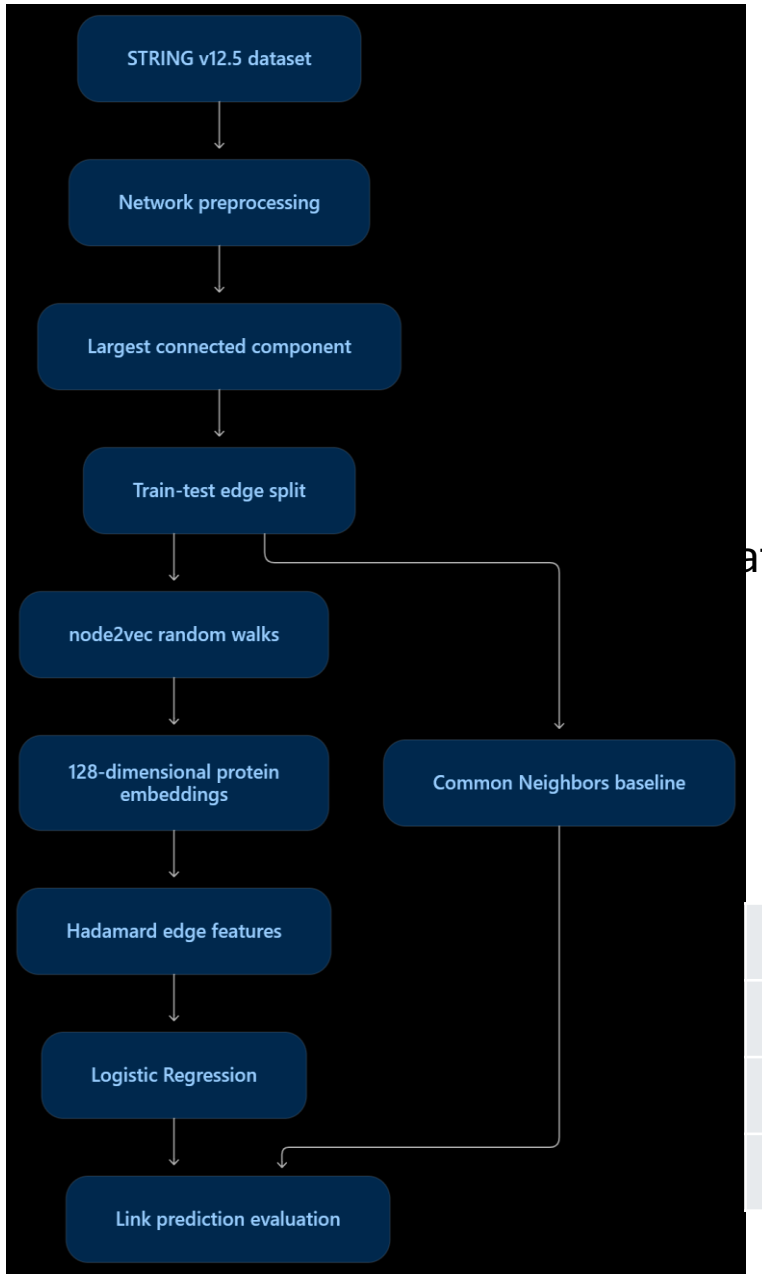
Only A–B interactions were retained.

Network construction:

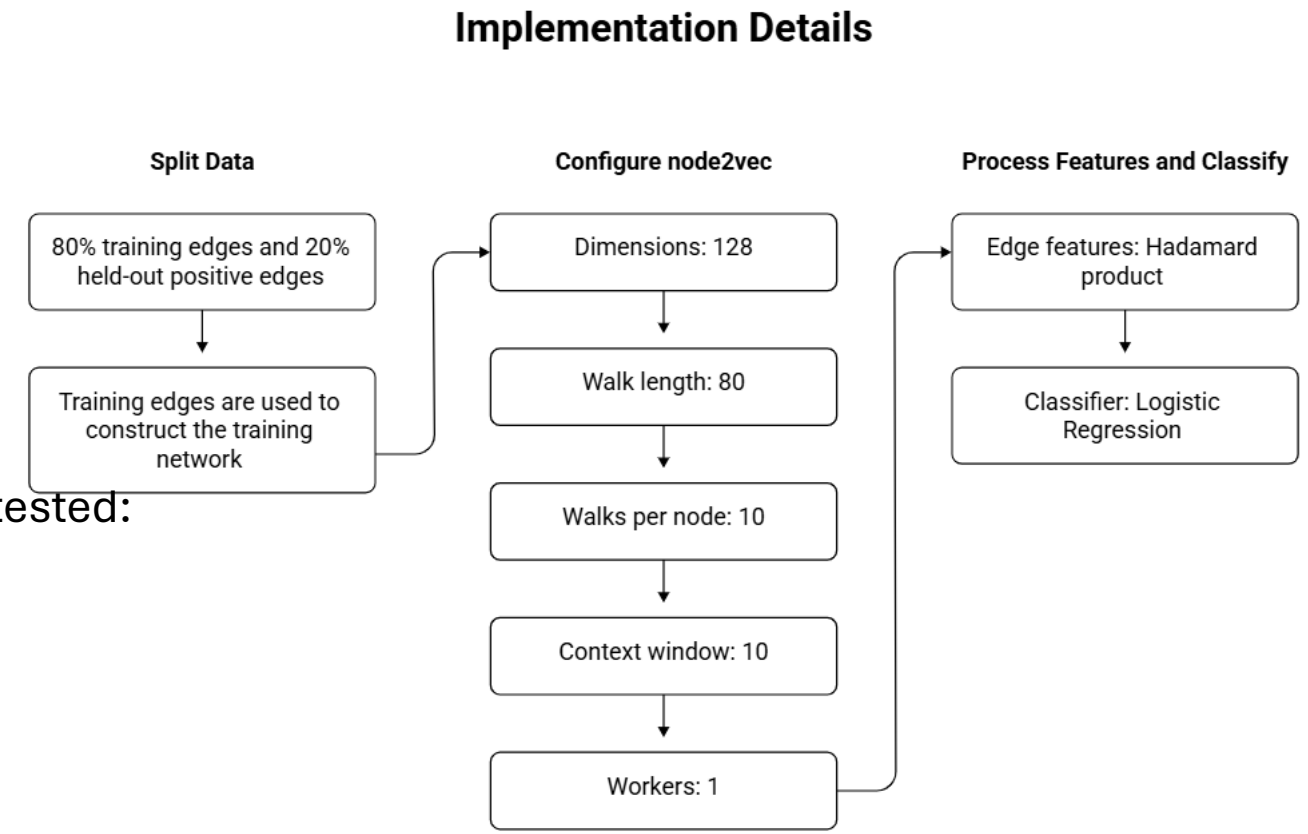


Stage	Nodes	Edges
Initial network	14,404	145,815
Largest connected component	13,996	145,498

Experimental Methodology



ations tested:

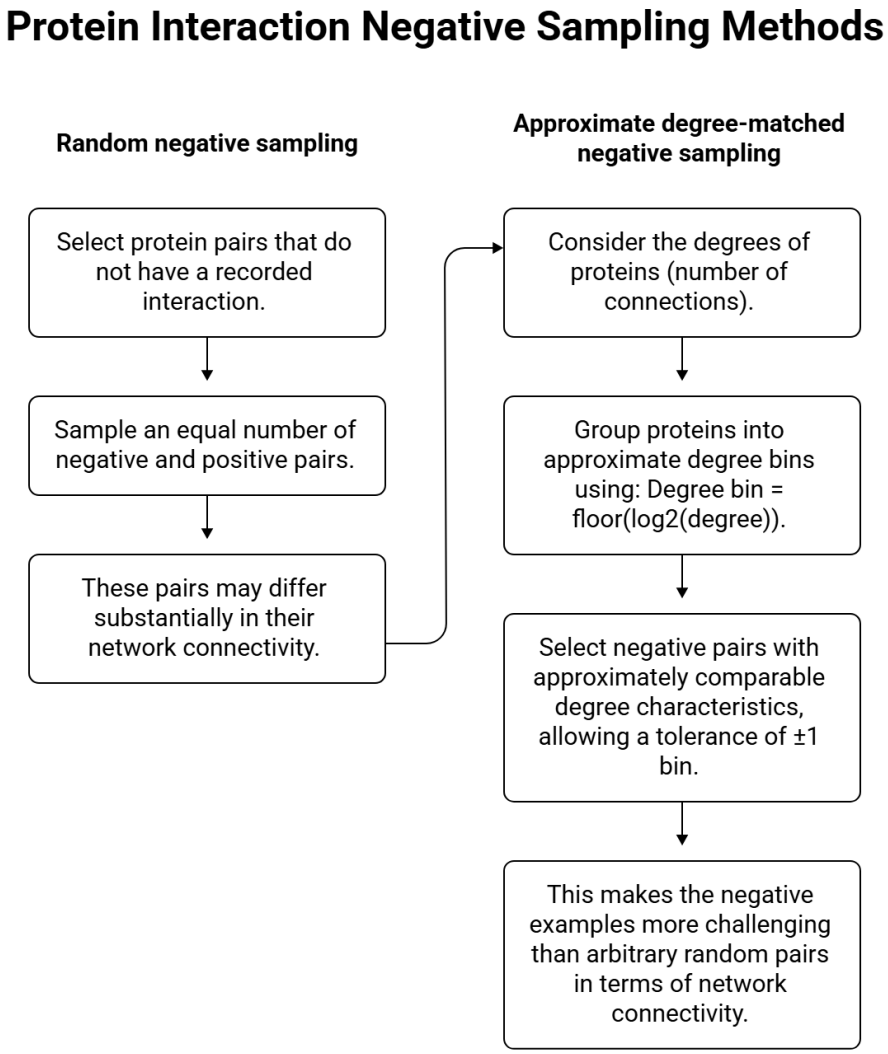


Configuration	p	q
Baseline	1	1
Local	1	2
Outward	1	0.5

The Change Beyond the Original Tutorial

What changed: **Negative-sampling strategy**

Instead of evaluating the model using only randomly selected negative protein pairs, here the comparison is for two strategies.



Results

Random negative sampling

Method	ROC-AUC
node2vec Baseline	0.9339
node2vec Local	0.9362
node2vec Outward	0.9324
Common Neighbors	0.9485

Degree-matched negative sampling

Method	ROC-AUC
node2vec Baseline	0.9371
node2vec Local	0.9379
node2vec Outward	0.9363
Common Neighbors	0.9209

Interpretation and Limitations

KEY OBSERVATION

Random negatives

Common Neighbors performed best

Degree-matched negatives

node2vec performed better than Common Neighbors

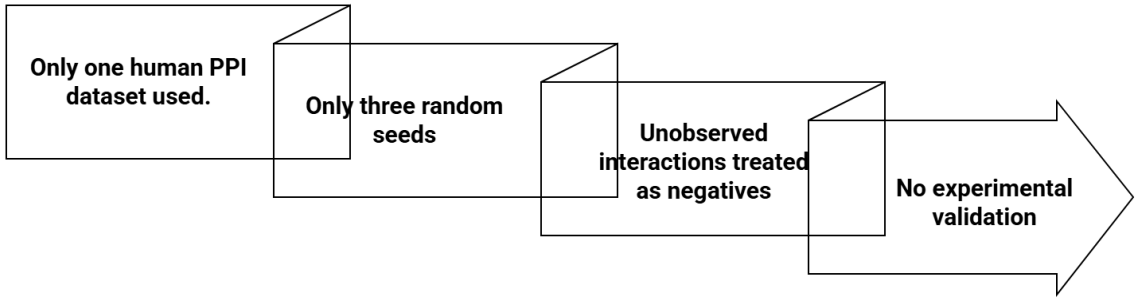
Local node2vec

highest mean ROC-AUC among node2vec configurations

Negative-sampling strategy

changed the relative performance of the methods

Limitations



Conclusion

- node2vec provides useful network representations for PPI link prediction.
- Evaluation strategy strongly influences model performance.
- Learned embeddings showed an advantage over Common Neighbors under degree-matched sampling.
- Further testing on additional datasets and experimental validation are needed.